

WELL SUMMARY & GENERAL INFORMATION

Well Name:	Duliajan-92 (DUL-92)	Field / Block:	Duliajan Field (Upper Assam Shelf)
Coordinates:	26.844° N, 95.328° E	Spud Date:	12-OCT-2022
Target Depth:	3,520 m (MD)	Completion Date:	18-JAN-2023
Primary Reservoir:	Barail Group Sandstone	Rig Assigned:	OIL Rig 14 (2000 HP)

INCIDENT CHRONOLOGY AT BARAIL GROUP SHALE (2,210 m)

On 14-NOV-2022 at 04:30 hrs IST, while tripping out of hole (POOH) at depth 2,210m in the Barail Group Shale section, the drill string experienced sudden mechanical entrapment (Stuck Pipe). Top drive rotational torque spiked from 12 kN·m to 28 kN·m, and overpull reached the maximum allowable limit of 45 tonnes without drill string movement.

■ **HAZARD CLASSIFICATION:** *Reactive Smectite/Illite Shale Hydration & Mechanical Pipe Encrustation.*

GEOLOGICAL & DRILLING PARAMETERS AT INCIDENT TIME

Parameter	Observed Value	Normal Baseline	Anomaly Deviation
Depth Interval	2,210.00 m - 2,214.50 m	2,150 m - 2,500 m	Barail Shale Zone
Mud Weight	1.16 g/cc (Water-Based)	1.24 g/cc EMW	-0.08 g/cc Underbalance
Gamma Ray (GR)	118 API	45 - 60 API	High Shale Content
Standpipe Pressure	2,850 psi	2,100 psi	+750 psi (Pack-off signal)
Rotational Torque	28.4 kN·m	11.5 kN·m	+16.9 kN·m (Entrapment Spike)
Shale Hydration Index	High Smectite (42%)	< 15%	Severe swelling hazard

APPLIED MITIGATION RECIPE & RECOVERY PROCEDURE

- Glycol Spotting Pill:** Mixed and pumped a 50 bbl specialized Poly-Glycol / High-Lubricity Spotting Pill into the annulus at 2,210m to dehydrate swollen shales.
- Soak Duration:** Allowed spotting pill to soak around BHA for 4 hours while maintaining 10 tonnes tension and working jars in downward direction.
- Mud System Inhibitor Adjustment:** Inhibited active mud system by raising KCl concentration to 8% and adjusting glycol content to 4% v/v.
- String Recovery:** Drill string freed at 09:15 hrs. Swept hole with 30 bbl high-viscosity nut-plug sweep and conditioned mud density to 1.25 g/cc.

TOTAL NON-PRODUCTIVE TIME (NPT): 36.5 Hours | **ESTIMATED FINANCIAL IMPACT:** ■45.6 Lakhs